

CML — Treatment (Tyrosine Kinase Inhibitors)



1. Imatinib (1st gen)

WHAT YOU SEE

IMATINIB banner with sword, 1x potency, ALL PHASES, ~\$500/yr

WHAT IT ENCODES

Imatinib is the first-generation ATP-competitive BCR-ABL1 inhibitor (IRIS trial, 2001) that transformed CML into a chronic disease. It is now generic and inexpensive, well tolerated, and a standard first-line option for low/intermediate-risk chronic-phase CML when survival is the goal. Also inhibits KIT and PDGFR (hence its use in GIST).

BOARD TRAP

Imatinib side effects: periorbital edema/fluid retention, muscle cramps, rash, cytopenias. It is slower to deep response than 2nd-gen agents — fine for survival, less ideal if aiming for fast deep response/TFR.

2. ATP-competitive target

WHAT YOU SEE

A red-and-white bullseye ATP target painted on the chest of the central wooden mannequin — the ATP-binding pocket every TKI aims at

WHAT IT ENCODES

Imatinib, dasatinib, nilotinib, and bosutinib are all ATP-competitive inhibitors that occupy the ATP-binding pocket of the BCR-ABL1 kinase, blocking phosphorylation of downstream substrates. This shared mechanism is why a single pocket mutation (T315I) can confer cross-resistance to all of them.

BOARD TRAP

Because these four bind the ATP pocket, the T315I 'gatekeeper' mutation blocks ALL of them — only ponatinib (ATP-pocket but T315I-active) and asciminib (allosteric/myristoyl) retain activity.

3. Dasatinib

WHAT YOU SEE

DASATINIB shelf, 300x potency, 100/50 mg, lungs icon

WHAT IT ENCODES

Dasatinib is a highly potent (~300x imatinib) 2nd-generation TKI, once daily, that also inhibits SRC-family kinases and penetrates the CNS (useful in lymphoid blast crisis/CNS disease). Achieves faster, deeper responses than imatinib.

BOARD TRAP

Dasatinib's signature toxicity is PLEURAL EFFUSION (and pulmonary arterial hypertension) — AVOID in patients with pre-existing lung disease/COPD or heart failure.

4. Dasatinib pleural effusion

WHAT YOU SEE

Lung icon with AVOID LUNG DX

WHAT IT ENCODES

Pleural effusion is the classic, dose-related dasatinib adverse effect (can occur years into therapy), along with pulmonary arterial hypertension. Managed by dose interruption/reduction ± diuretics/steroids; switch TKI if recurrent.

BOARD TRAP

New dyspnea/pleural effusion in a CML patient on a TKI → think DASATINIB. Avoid dasatinib if the patient has lung disease.

5. Nilotinib

WHAT YOU SEE

NILOTINIB shelf, 30x, 300 BID, AVOID DM/CVD

WHAT IT ENCODES

Nilotinib is a 2nd-generation TKI (~30x imatinib potency), dosed twice daily and must be taken on an empty stomach (food increases absorption → toxicity). Produces fast, deep molecular responses, good for pursuing TFR.

BOARD TRAP

Nilotinib causes HYPERGLYCEMIA, QT prolongation, and vascular/arterial occlusive events (PAD, MI, stroke) — AVOID in diabetics and patients with cardiovascular disease. Also pancreatitis/lipase elevation. Empty-stomach dosing + QT risk makes it the 'metabolic/cardiac' TKI.

6. Bosutinib

WHAT YOU SEE

BOSUTINIB shelf, 400 mg, GI/liver icons

WHAT IT ENCODES

Bosutinib is a 2nd-generation dual SRC/ABL inhibitor used mainly after failure or intolerance of other TKIs (and as a first-line option). It has relatively little cardiac/pulmonary toxicity, making it attractive when those organ systems are a concern.

BOARD TRAP

Bosutinib's signature toxicities are DIARRHEA (very common, usually early/self-limited) and HEPATOTOXICITY (transaminitis) — caution with hepatic impairment. Choose it to avoid the cardiac/pulmonary toxicities of nilotinib/dasatinib.

7. Ponatinib

WHAT YOU SEE

PONATINIB shelf, 45→15 mg, HEAVY HAZARDS, pancreas/fire

WHAT IT ENCODES

Ponatinib is the 3rd-generation TKI specifically active against the T315I gatekeeper mutation (and other resistant clones). Reserved for T315I-positive disease or failure of ≥2 prior TKIs. Often dose-reduced (45→15 mg) once a response is achieved to limit toxicity.

BOARD TRAP

Ponatinib carries the HIGHEST risk of arterial thrombotic/cardiovascular events (MI, stroke, PAD) and causes PANCREATITIS — reserve for T315I or multi-TKI failure, and reduce dose after response.

8. T315I gatekeeper mutation

WHAT YOU SEE

An armored knight figure crouched behind a riot shield stamped 'T315I' — the gatekeeper mutation shields the pocket from ATP-competitive TKIs

WHAT IT ENCODES

T315I is the 'gatekeeper' kinase-domain mutation: a threonine-to-isoleucine substitution that sterically blocks ATP-competitive TKIs from binding. It confers resistance to imatinib and all standard 2nd-generation TKIs and is the most common cause of pan-TKI resistance.

BOARD TRAP

T315I → use PONATINIB or ASCIMINIB; it does NOT respond to imatinib, dasatinib, nilotinib, or bosutinib. (Contrast: Y253H/E255K → dasatinib; V299L/F317L → nilotinib.)

9. Highest CV risk flag

WHAT YOU SEE

HIGHEST CV banner near ponatinib

WHAT IT ENCODES

Ponatinib has the highest cardiovascular/arterial occlusive toxicity of any TKI — a black-box concern. Mitigation includes aggressive cardiovascular risk-factor control and dose reduction (to 15 mg) once a molecular response is reached.

BOARD TRAP

If a board stem describes a CML patient who develops MI/stroke/limb ischemia on therapy, think ponatinib (highest CV risk) — or nilotinib (vascular events). Imatinib has the cleanest cardiovascular profile.

10. Asciminib (STAMP)

WHAT YOU SEE

ASCIMINIB boomerang, 40 BID / 200 BID T315I, STAMP

WHAT IT ENCODES

Asciminib is the first allosteric BCR-ABL1 inhibitor — a STAMP (Specifically Targeting the ABL Myristoyl Pocket) agent. By binding the myristoyl pocket rather than the ATP site, it works against many TKI-resistant clones, including T315I at higher dose (200 mg BID vs 40 mg BID standard). Used after ≥2 prior TKIs.

BOARD TRAP

Asciminib is NOT ATP-competitive — it targets the myristoyl (allosteric) pocket, so it retains activity against T315I and many ATP-pocket mutations that defeat conventional TKIs.

11. Omacetaxine

WHAT YOU SEE

OMACETAXINE bench, 'NOT A TKI', after ≥2 TKI failure

WHAT IT ENCODES

Omacetaxine (homoharringtonine) is a protein-synthesis inhibitor — NOT a tyrosine kinase inhibitor — given subcutaneously for chronic/accelerated CML after failure of ≥2 TKIs. It acts independently of the kinase domain, so it can work even with T315I.

BOARD TRAP

Omacetaxine is NOT a TKI (mechanism = protein synthesis inhibition); it is a multi-TKI-failure salvage option, including in T315I disease.

12. Survival = generic imatinib

WHAT YOU SEE

Bottom plaque: SURVIVAL -> IMATINIB GENERIC

WHAT IT ENCODES

When the treatment goal is overall survival alone, generic imatinib is sufficient first-line for most chronic-phase patients and is the most cost-effective choice — long-term survival is comparable across first-line TKIs.

BOARD TRAP

Overall survival is similar across first-line TKIs; the advantage of 2nd-gen agents is faster/deeper response, not necessarily longer life. So 'goal = survival' → generic imatinib is a perfectly correct answer.

13. High risk/TFR -> 2nd gen

WHAT YOU SEE

Bottom plaque: TFR/HIGH RISK -> 2nd GEN

WHAT IT ENCODES

For high-risk disease (high Sokal/ELTS score, young patients, those prioritizing treatment-free remission) start a 2nd-generation TKI (dasatinib/nilotinib/bosutinib) first-line, because they produce faster and deeper molecular responses — the prerequisite for attempting TFR.

BOARD TRAP

Choose the first-line TKI by GOAL and COMORBIDITY: survival/elderly/low-risk → imatinib; high-risk or aiming for TFR → 2nd-gen, but match the agent to the patient (avoid dasatinib in lung disease, nilotinib in DM/CVD, bosutinib in GI/liver disease).